

Design | 40 MIN READ | FEATURES

Design Document: Play With Fire

For today's exclusive Gamasutra feature, International Hobo's Chris Bateman provides a rare public look at an in-depth commercial PlayStation 2 game design document with *Fireball*, eventually released on PC as *Play With Fire* by Manifesto Games.



Chris Bateman

Blogger

February 21, 2007



[In the spirit of community, and for the sake of education, [International Hobo's](#) Chris Bateman has provided a rare public look at an in-depth commercial game design document. The game, here tentatively titled *Fireball* and targeted for the PlayStation 2, was released for the PC by Manifesto Games with the new title of [Play With Fire](#).



A second document was also provided, the *Fireball Field Design Guide*, which can be downloaded in *.doc format [here](#).]

Fireball

火玉 Hidama

1. Burn It Down



Example level from Play With Fire

1.1. Overview

Fireball is a budget game for PS2. The player controls a ball of fire, and traverses a landscape made of blocks of different materials. As the player sets fire to these blocks, they grow hotter, and can set fire to more and more different types of blocks. The fireball the player controls can also rise up in height and the hotter the player gets, the higher they can jump in this fashion.

On each field (level) the player has an ultimate goal of igniting the torch (brazier) and thus clearing the field – but the torch is generally positioned at a high point and out of reach. The player must use a combination of platform skills and dynamic environmental features (for instance, by burning the supports under the torch down to the ground) in order to clear the field.

Simple, clean cut graphics and controls combine to give an easy to learn but engaging play experience.

1.2. Vision Statement

The player should be delivered the following experiences:

- **Effortless** play originating from a simple control scheme.
- **Unique** experience – the only game to be based around setting fires.
- **Varied** solutions to the mini-puzzles as the player works out the best places to start fires.
- **Exploration** of small environments.

1.3. Branding Choices

There are two options for the brand image for the game, and which is used depends entirely upon the needs of the choice of the publishing partners:

- **Fireball Brand:** this approach targets impulse purchasers and is suitable for where the primary sales channels for the game are in mass market catchment areas e.g. supermarkets. In this case, the game packaging should draw attention to the fact that it is **Easy to Play**, in order to appeal to the Casual audience, who typically finds most games are too hard to control.
- **Hidama Brand:** the Hidama (Japanese for 'fireball' or 'falling star') approach is targeting the Hardcore gaming community's desire for new and innovative product. A Japanese sub-title for the game serves to intrigue, and suggest that this may be an interesting and hitherto overlooked title (such as *Katamari Damacy*). The game packaging should draw attention to the games **Unique Gameplay**. This approach is best suited if the publisher is expecting to sell the game primarily through specialist game shops or online.

A hybrid approach may also be undertaken.



All the play in *Fireball* originates from three simple to implement core subsystems:

- **Avatar** concerns the player's ability to negotiate the landscape.
- **Temperature** concerns the ignition of blocks in the playing field, and how fire spreads between blocks.
- **Gravity** concerns the collapse of objects and blocks in the playing field as a result of fires.

We will discuss each in turn.

2.2. Avatar

2.2.1 Overview

The player's Avatar is a glowing ball of fire, considered to be 1 unit in diameter. The player's abilities are as follows:

- **Move** around the environment. The player turns left and right, and pushes forward to move (relative controls).
- **Jump** up to a (relative) height determined by the heat of the ball. The characteristics of this jump are that the player rises rapidly up to their maximum (relative) height, and then slowly descend.
- **Burning** blocks is achieved simply by pushing into them. If the player is just hot enough to ignite a block, they will need to push into the block for a short while to start a fire – but if they are considerably hotter, fires will start just by them passing by.
- **Slamming** can be done after the player has jumped – it causes the fireball to come crashing down to the first surface beneath them, igniting fires quickly in a wide area of impact where they come down. These fires are slightly hotter than the player's default temperature.



These are all the player's abilities.

2.2.2 Avatar Temperature

The basis of fire starting rests in a simple system of temperature based upon colours. The avatar increases in heat permanently when it touches a burning block that is hotter than the avatar's current temperature.

Colour	Jump Height	Description
1: Yellow	+2 units	Yellow flames; bunsen burner style
2: Orange	+4 units	Bright orange flames.
3: Red	+8 units	Glowing red with heat haze.
4: Blue	+12 units	Blue-white flame, like a blowtorch.
5: White	+16 units	Bright white glow – very bright.

2.3. Controls



Basic Controls



Jump

The fireball jumps up to its maximum height, then begins to drift slowly down towards the ground.



Slam

Crash down to ground rapidly and then explode, igniting nearby blocks. If already on the ground, just explodes.



Jump and Top Down View

The fireball jumps, but the camera view tilts to give a top down view. Press again to cancel top down view. (Toggles top down view).



Pause/Map



Reset Field

Hold for 0.5 seconds to begin the current field again

Advanced Controls



Roll Left

Move Sideways to Left i.e. strafe left



Roll Right

Move Sideways to Right i.e. strafe right



Turn Left

Turn 90 degrees left



Turn Right

Turn 90 degrees right

2.3.1 Jump Profile

The following is a description of the jump profile for the fireball:

- **Start** at any given height. This may be the ground (0 height) or on a platform (>0 height).

- **Drifting** begins immediately. The player descends at the rate of about 1 unit per second.
At any point, the player can use Triangle to toggle between a top down view and a regular view.
- **Pick Target** by looking at the shadow of the fireball. This is easiest in the top down view. Regardless of lighting, the shadow of the fireball always shows precisely where the player will land if they press Slam.
- **Slam** by pressing the appropriate control. The fireball descends almost instantly to the shadow-point and explodes – possibly igniting everything at this point. The slam fires are at a temperature one higher than the avatar's current temperature.

Note that it is not possible to jump again once in the air – the fireball must be in contact with a block in order to jump.

Also note that there is no quick way to descend except for slamming.

2.3.2 Slam Profile

Whenever the player slams, they raise every block in a 3 unit spherical radius of the point of impact (or point of explosion if they were on the ground) up to one higher than their current temperature. (The colour of the explosion effect should correspond to the higher temperature).

If this temperature is high enough to ignite a block, the block begins burning.

2.4. The Player's Goal

The player's goal is always to move, burn and melt their way around the environment in order to reach the torch – a symbolic brazier item – which they ignite on contact.

As soon as this is done, the playing field is cleared.

Advanced players will attempt to clear the playing field in the shortest play time, and/or cause the biggest Chain – achieved by having large number of blocks burning at the same time.

The player scores more for big Chains, and their score contributes to an overall progress mechanism which unlocks new collections of levels and simple cosmetic rewards.

2.5. Gaining Temperature

Increasing avatar temperature is (usually) one of the key steps the player undertakes towards their goal. To overcome various barriers, the player must often become hotter. This is achieved by touching a block that is burning at a higher temperature than the avatar. This can happen:

- If the avatar slams a material they could not usually burn in order to ignite it. This would immediately 'level up' the avatar to the new temperature
- If the avatar touches a material burning at a higher temperature. Such a block could be ignited by the heat of other blocks the player has ignited.

Once the avatar temperature increases, it is permanently at the higher degree for the rest of the current field.

2.6. Chains

2.6.1 Overview

Although it is not the player's stated goal to get a big Chain, the game is structured to imply that ending the level is not really victory – getting a large Chain is victory. This allows players to find their own level of challenge... If just reaching the brazier is challenging, the player will feel satisfied by this goal. If not, scoring a high Chain will seem like the goal, with the brazier merely being the end of the current field.



The size of the current Chain is equal to the number of blocks currently burning.

2.6.3 Font Size of the Chain Counter

The size of the text which displays the current Chain count depends upon whether or not the Chain number has been growing or receding. A strict rubric defines this relationship:

- When no blocks are burning, no Chain is displayed.
- The first time a block is ignited, the font size is "100%". (All other sizes are relative to this size). This text will remain displayed for 10 seconds, gradually fading out if it is not 'replaced' with a new Chain value.
- The next time that the Chain value is updated (which may happen several times a second), if the Chain value is larger, the font size should be increased by the amount the chain is larger.

Font Size = 100 + [Chain value] percentiles

- If the Chain value remains the same, the font size remains the same.
- If the Chain value decreases, the font size decreases accordingly.
- A maximum font size should be specified, based upon the largest size that can be displayed without obscuring the player's view.



If the text can only be displayed at certain discrete sizes, then the Font Size variable should be used to test whether or not the threshold for the next font size applies, rather than as a direct scaling factor.

2.7. Field Reset

The fields in *Fireball* often present mini-puzzles. Sometimes, these puzzles can end up in states from which the player cannot reach the goal. For this reason, a control is provided (Select) to quickly and easily restart the field *ideally without any noticeable loading*.

It is expected that normal play will involve a fair amount of resetting.

Select must be held for at least 0.5 seconds to reset the field; this prevents the field being restarted by accident. The pause menu will carry the text "Hold Select to reset the field."

3. Environment

3.1. Components

3.1.1 Blocks

The environment is entirely constructed out of 1 unit cubes (actually about 4 m per side, therefore 1 unit = 4 m). These cubes have different colours, and are textured to resemble specific materials.



3.1.2 Objects

Objects are simply clusters of blocks. For instance, a vertical column of ten blocks is an object. Four such columns with a flat plane of blocks across the top is a 'Table' object.

Clearly, because objects are made of cubic blocks, they are abstract in nature, but the player will still be able to make out what these objects represent.

3.2. Gravity

Gravity always pulls blocks and objects downwards. The gravity value is 10 units per second per second, with a terminal velocity of 5 units per second.

Objects fall vertically downwards until any one of their constituent blocks hits another block underneath – then they stop. This applies, even if the resultant structure would look physically impossible. As long as there is a block underneath, the object will balance where it is.

3.3. Types of Block

The following are the types of Block that all objects are made of:

	Colour				
1: Leaf	Green	No	1: Yellow Hot	10 seconds	0.1 seconds*
2: Wood	Brown	No	2: Orange Hot	15 seconds	1 second
3: Coal	Black	No	3: Red Hot	60 seconds	1 second
4: Plastic	Pink	2: Orange Hot	3: Red Hot	15 seconds	1 second
5: Metal	Blue	4: Blue Hot	5: White Hot	90 seconds	1 second
6: Stone	Grey	5: White Hot	No	–	–
7: Fire	Red	No	No	–	–



*In fact, ignites slower except when exposed to extreme heat.

Note

No block may ever be at a heat level higher than that shown in its Burn column or Melt column (whichever is higher).

The following table shows the tints of blocks when they are melted or burning:

Block Type	Block Colour	Melted Texture	Meted Tint	Burning Texture	Burning Tint
Leaf	Green	–	–	Burning Leaves	Yellow
Wood	Brown	–	–	Burning Wood	Orange
Coal	Black	–	–	Hot Coals	Red
Plastic	Orange	Molten Plastic	Orange	Burning Plastic	Red
Metal	Blue	Molten Metal	Blue	White-hot Metal	White
Stone	Grey	Lava	White	–	–

3.4. Burning

3.4.1 Ignition

The temperature at any point in the game field (for the purposes of checking for ignition) is based upon the temperatures of the surrounding blocks. The process of determining if any given block ignites is as follows:

- Burning blocks within 1 unit (including diagonals which are technically 1.4 units away) provide 0.5 of their burn temperature.
- Burning blocks within 2 units (including diagonals which are technically 2.8 units away) provide 0.1 of their burn temperature.
- Burning blocks within 3 units (including diagonals which are technically 4.2 units away) provide 0.02 of their burn temperature.

Blocks are only counted as Burning if they have been on fire for at least as long as the ignition time for their relevant material.

If the temperature at any given point is higher than the required temperature for ignition for any given block, it catches fire (and then burns at the same temperature as its ignition).

Note

The three parametric values above are the temperature radiation coefficients (TRC1, TRC2 and TRC3). The values given should be considered default values – tweaking will be required.



Ignition takes 1 second. During this time, the block is considered to be at 0 temperature (not at its burning temperature). Only once it is alight does it take on its new temperature.

For example, consider the following example (in 2 dimensions):

0 seconds: The tree begins burning at the centre top (as indicated by the '1' – which is the temperature for burning leaves):

			1						

1 second: The neighbouring blocks immediately ignite. The blocks around have temperature determined only by the originally burning block:

		(1)	1	(1)					
	0.1	(1)	(1)	(1)	0.1				

2 seconds: The fire spreads. Notice how the 'bow wave' of the fire has risen from 0.1 to 0.36.

		1	1	1					
	(1)	1	1	1	(1)				
	(1)	(1)	(1)	(1)	(1)				
		0.36	0.36	0.36					
			0.06						



3 seconds: The leaves are fully alight. The wood is still safe.

		1	1	1					
	1	1	1	1	1				
	1	1	1	1	1				
		(1)	(1)	(1)					
			0.44						
			0.06						

4 seconds: The temperature at the top of the trunk is now effectively 2.1. This ignites the wood (which burns at Temperature 2).

		1	1	1					
	1	1	1	1	1				
	1	1	1	1	1				
		1	1	1					
			(2.1)						

5 seconds: The temperature at the top of the trunk is now effectively 2.1. This ignites the wood (which burns at Temperature 2).

		1	1	1					
	1	1	1	1	1				
	1	1	1	1	1				
		1	1	1					
			2						
			(2)						



6 seconds: The entire tree is burning.

		1	1	1					
	1	1	1	1	1				
	1	1	1	1	1				
		1	1	1					
			2						
			2						

Notice that there is a maximum temperature that a given material can contribute to another material. Imagine that there was metal in the centre of the tree:

		1	1	1					
	1	1	4.2	1	1				
	1	1	4.2	1	1				
		1	1	1					

The temperature from the burning leaves would not be enough to ignite the metal.

Any blocks or objects that were resting upon that block then fall down. Thus the shape of the world changes over time as more objects and blocks burn to nothing.

3.4.3 Polling Blocks (Checking for Ignition)

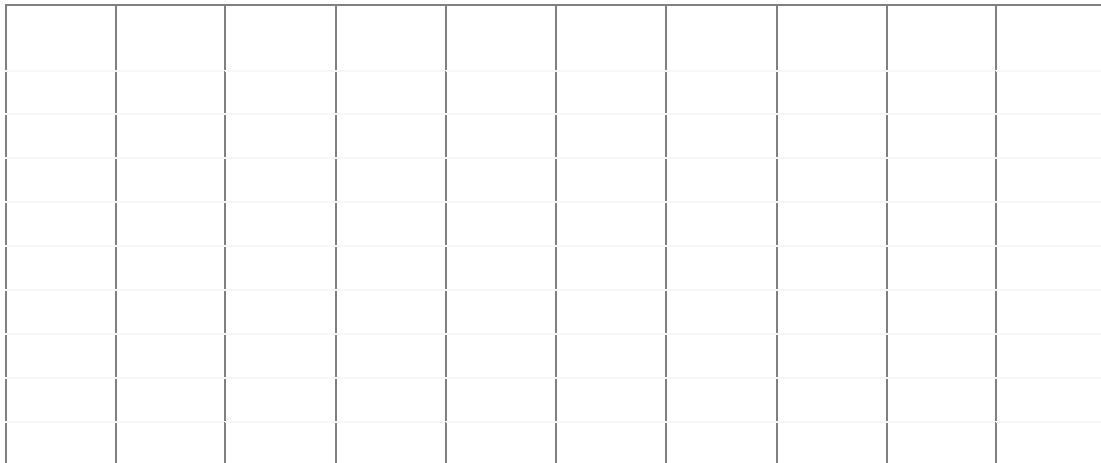
The game engine should poll the blocks of the world at the fastest rate feasible without causing slowdown – blocks near to the player should be polled in as close to real time as possible, whilst blocks far from the player are lower priority. But within an approximately ten second cycle, all blocks in the game world should be polled at least once to determine if they start any fires nearby.

3.5. Melting

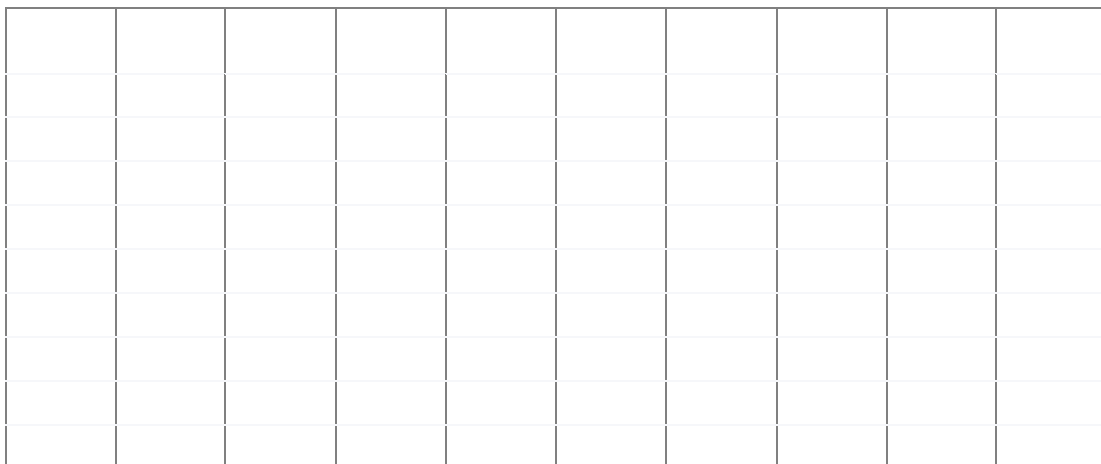
3.5.1 Collapsing

When an object melts, it means that it ceases to be treated as a composite object of blocks and instead becomes a collection of unconnected blocks.

Consider the following diagrams (in two dimensions for simplicity):



This shows a Tree object, made of a set of Leaf blocks on top of a Wood block pole. It is standing on a Plastic block table.



The plastic table melts. The tree falls into the table (which has effectively inverted itself by melting) but gets stuck on the edge of one of the 'legs'.

To make melting objects look like they are melting, the gravity effect on the blocks of an object which has melted should be variable.

Set the gravity for each block to be between 1 and 5 units per second per second, with a terminal velocity of 5 units per second. This will give the impression of melting.

3.5.3 Pushing Through

The other effect of melting temperatures is that the player can push through a barrier if they are hot enough to melt.

For instance, if the player is Orange Hot (2) and there is a 'wall' of Plastic Blocks ahead of them, they are hot enough to melt these blocks and they can push through. The player simply moves up to the block, and moves through.

Their movement is slowed to 75% of normal when they push through.

After pushing through, the block where the fireball passed is removed from the field, exactly as if it had been burned out.



4. Structure

4.1. Overview

The game is divided into **field lists** or **Quests** which consist (in general) of six fields. (The term Quest is used in game documentation, but in the design document the term Field List is used). The player must play through all six fields to complete a field list. However, they can exit their current set of six fields by pressing start and choosing exit from a pause menu.



The game is made of three paths – *Fun*, *Puzzle* and *Challenge*, which are accessed from a central Hub screen. Each path consists of 10 field lists, for a total of 60 fields each or 180 overall. There may, however, be up to 40% reuse of material between the different paths, meaning the game will actually consist of between 108 and 180 fields.

Medals are earned for hitting certain combo targets in field lists. These unlock the field lists from the Challenge path.

Additionally, every block the player burns earns 1 point of Ash. A running Ash total is used to unlock additional materials in the Fun path.

4.2. Ash

Ash is earned for each block burned. It is used to progress along the Fun path. The target Ash values for the Fun path will rise exponentially.

4.3. Rewards

4.3.1 Medals

Every field list has a target Chain total associated with it. This relates to medals as follows:

- If the sum of all Chains scored for all six fields is equal to or exceeds the target Chain for the field list, a **Gold Medal** is awarded. Whenever the avatar is close enough to the Quest block in the Hub for that field list, a gold medal icon is shown. When a Gold Medal is awarded, a Silver medal is automatically awarded as well.
- If the total Chain scored is between 50 and 99% of the Gold Medal score, a **Silver Medal** is awarded. This is also shown as an icon when the avatar is close to the relevant Quest block in the Hub.

Whenever the player completes a field list, they are shown the amount of Ash earned. Plus:

- If they scored a Silver Medal, this icon is shown, and the Ash total is multiplied by 2 (x2 is shown).

4.3.2 Summary

The following summarises the ‘prizes’ that can be won:

Medal	Combo Target Percentage?	All Fields Ash Clear?
Silver	50–99%	No
Gold	100%	No

4.4. Front End



4.4.1 Overview

The game front end operates using the main game engine. The player moves around between objects (which are kept well-spaced), and when near an object, text appears indicating what will happen if they burn that object.

This saves developing a separate front end.

4.4.2 Quest Blocks (Hyperobjects)

The Hub field consists of a number of blocks, which are regular game objects – but that cause fields or Quests to be run when they are burned by the player. In effect, each block is a menu option that the player selects by moving to it and burning it.

The blocks displayed in the Hub will be either Leaf, Wood or Stone.

4.4.3 Unlocked Quests

Any Quest that has been unlocked is displayed as a single Leaf Block.

This is exactly as is currently implemented

4.4.4 Completed Quests

Any Quest that has been completed is displayed as a single Wood Block.

This means:

- Players won't trigger these by mistake, as they must Slam to burn a Wood block
- Players can immediately see which Quests they have completed

4.4.5 Next Quests

The next Quest to be unlocked in each Path is displayed as a single Stone Block. Clearly, this cannot be burned, and so the Quest cannot be played.

Above the block is the description of how the Quest is unlocked – see Section 4.8 for details.

4.4.6 Ash Total

While on the Hub field, a large, graphically pleasing display of the players total Ash should be shown at the bottom of the screen.

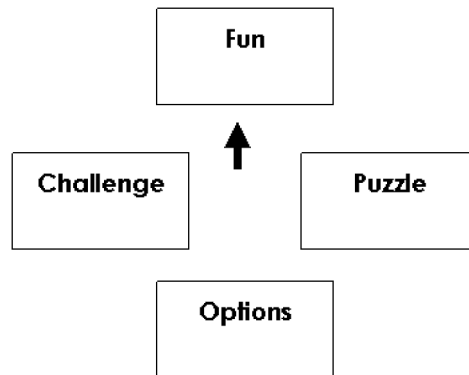
4.4.7 The Hub

The main game functionality is accessed from a game field which is the hub map. It consists of a set of blocks that fit into three paths:

arcade control skills to master. The Puzzle path advances from the completion of Quests.

- **Challenge Path:** this consists of challenging fields which may require good reactions and arcade control skills to master. The hardest fields are in this path. This path will advance through Medals.

All three paths are arranged to radiate from a central point. The Options area is in the fourth position, as follows:



The player begins facing the first object in the Fun Path.

4.5. Auto-saving

The game automatically saves after completing a field list. Any completed field lists are recorded, as are any medals awarded, plus the player's Ash total and the highest combos scored for each field list.

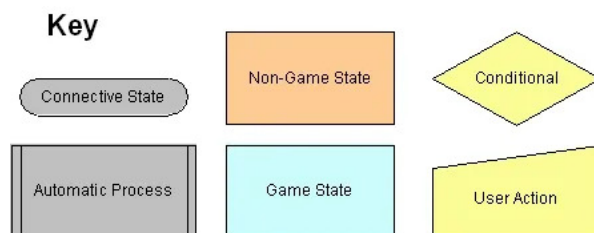
No manual save is provided.

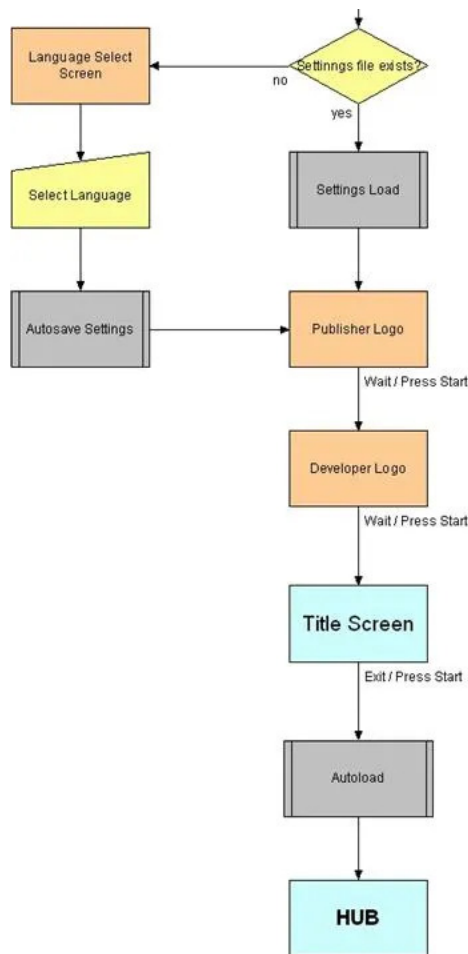
If the player exits a field list, the game still saves – any Ash earned is still 'banked'.

4.6. High Level States

4.6.1 Key

The following is the key to the state diagrams used in this document:

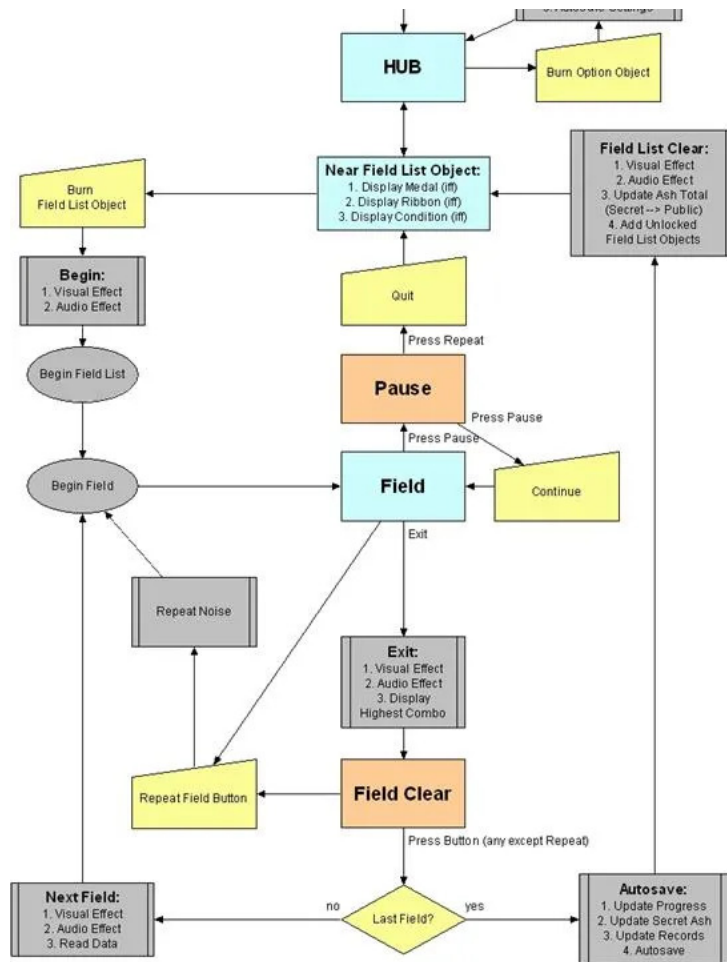




**Start Up
State Diagram
for Fire ball**

The states in this diagram are as follows:

- **Start:** game begins
- **Settings file exists?** Check for settings file; action varies according to whether this is present.
- **Language Select Screen:** if there is no settings file, a language select screen appears displaying the flags of all the countries whose languages are supported in a vertical column.
- **Select Language:** The player selects their language by selecting a flag and pressing X or Start.
- **Autosave Settings:** saves the language setting to a (new) settings file.
- **Settings Load:** alternatively, if a settings file exists it is loaded.
- **Publisher Logo:** In the best case, it would be nice if this could be “painted” onto a wall of blocks that could then be burned by the player. However, a single static image flash up can be used instead. In *either case*, pressing Start (joypad) or Esc (keyboard) should advance the screen to the next screen.
- **Developer Logo:** International Hobo Ltd will use the Fantasy Labs logo as their developer logo for this game. As above, in the best case, it would be nice if this could be “painted” onto a wall of blocks that could then be burned by the player. However, a single static image flash up can be used instead.
- **Title Screen:** After the publisher logos, the player should be dropped into the Title Screen field (Title.Leasy.spiral.fsf). They can either move into the exit, or press Start (joypad) or Esc (keyboard) to move to the Hub.
- **Autoload:** after the Title Screen, the save file is automatically loaded, restoring any game progress, medals and Ash total.



**High Level
State Diagram
for Fireball**

The states in this diagram are as follows:

- **Start up:** see the Start Up Sequence flow diagram.
- **Hub:** the game hub. If the player goes near a Field List Object, becomes slightly different state (see below).
- **Hub – Near Field List Object:** in this state, the game should display any Medal and/or Ribbon for that Field List. If the object has not yet been unlocked, the condition for unlocking it is displayed (see Path Progression, below).
 - Burn Option Object
 - Change Option/Replace Option Object
- **Burn Field List Object:** this is the trigger for starting a field list.
- **Begin:** play visual and audio effect to show start of field list (this is the same as the visual and audio effect at the beginning of any field – the *Begin Fanfare*).
- **Begin Field List & Begin Field:** these are defined as states for reference only.
- **Field:** the game.
- **Pause:** pause screen; accessed when player presses Start or Esc.
- **Continue:** return to the game.
- **Quit:** return to the Hub.

as the highest combo value.

- **Last Field?** Conditional; if it is the last field, go to Autosave, else Next field.
- **Next Field:** play the sounds and visual effects for starting a new field, then start the next field.
- **Autosave:** save progress. The 'Secret Ash' referred to here is the *actual* total amount of Ash the player has earned. This is different from the Total Ash value which the player sees as the Total Ash is only updated when returning to the Hub (just after the autosave).
- **Field List Clear:** play the appropriate effect for any Medal earned, and update the Total Ash so that it matches the Secret Ash (i.e. show Ash ticking up until it is equal to Secret Ash. Secret Ash is the real amount that the player has burned, Total Ash is just what the player sees at any given time.

4.7. Overlays

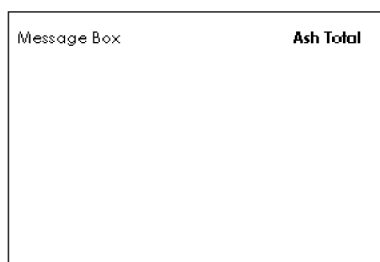
4.7.1 Introduction

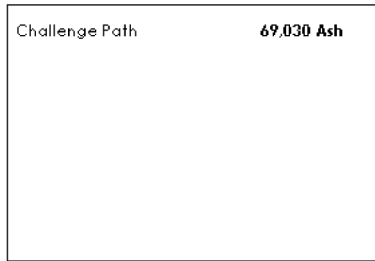
This section describes the HUD-style overlays used in the game. The components of these overlays are as follows:



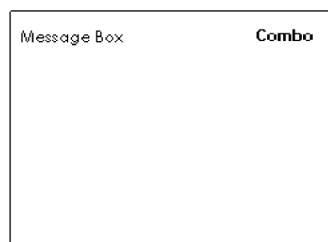
- **Message Bar:** this is a 'box' in the top left hand corner which displays information in text. It should *not* have a line around it; the text should be shown on its own.
All overlays
- **Ash Total:** the total Ash the player has accumulated so far. It is suffixed by the word **Ash** to clearly denote it's meaning.
Hub overlay
- **Ash Score / Ash Target:** the amount of Ash earned in the current field and the maximum amount of Ash that can be earned.
Pause overlay
- **Combo:** the current combo.
Game overlay
- **Highest Combo:** the highest combo scored so far
Pause overlay
- **Audio Player:** this displays the current track. The player can push left and right to change the track.
Pause overlay
- **Quit/Continue:** this displays the instructions to the player to either quit the current field list (and return to the Hub) or continue with the current field.
Pause overlay

4.7.2 Hub Overlay

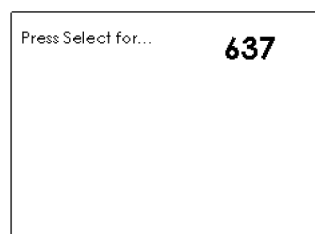




4.7.3 Game Overlay



The following is the Game overlay:



Example:

4.7.4 Pause Overlay

The following is the Pause overlay:



Example:

4.7.5 Message Box

The message box displays up to 32 characters.

The following table shows the behaviours of the Message Box:

Overlay/State	Status	IF (Condition)...	Message
Hub	In Fun Area		Fun Path
	In Puzzle Area		Puzzle Path
	In Challenge Area		Challenge Path
	In Options Area		Options
	Default		Find something to burn...
	By Specific Option		[Option Message]
	By Field List Object	Object Unlocked	Burn to attempt this Quest
Game		Object Not Unlocked	[Condition Message]
	First 3 seconds of play	Field Titles Enabled	Field Name
		Field Titles Disabled	null
	Default	Select Reminder Enabled	Press Select to Restart
		Select Reminder Disabled	null

4.7.6 Audio Player

On the Pause screen will be displayed the controls for the audio player:

- The current track name is shown between two enclosing triangles e.g.
◀ Red Birds ▶

- By default, the music plays in sequence, however, a "Random" option can be included in the track list to enable them to play in a random sequence.



This will remain based upon Ash.

A new Ash value sequence is included later in this document.

4.8.2 Challenge Path

This will still be based upon Medals, but instead of the incoherent Silver and Gold threads currently used, we will switch to a single Medal count:

- Silver Medals count as 1 medal
- Gold Medals count as 1 medal, but if the Silver was not previously earned it is counted too (so that if you have the Gold Medal for a Quest, you have 2 medals – 1 Silver and 1 Gold)

A new sequence is included later in this document.

4.8.3 Puzzle Path

To progress in the Puzzle path the player simply has to beat the Puzzle Quests in sequence.



4.8.4 Credits

The credits should now play when *all 10* of the main stages (1 to 10) in a Path are completed for any Path – e.g. if Fun 01 through to Fun 10 is completed, whenever the last of these is completed, the credits should play.

4.8.5 Conditions

The following table describes the progression of the three Paths:

Stage	Fun Path	Puzzle Path	Challenge Path
01	Always Available	Always Available	Always Available
02	500 Ash	Clear Puzzle 1	1 Medal
03	1,000 Ash	Clear Puzzle 2	2 Medals
04	5,000 Ash	Clear Puzzle 3	4 Medals
05	10,000 Ash	Clear Puzzle 4	6 Medals
06	15,000 Ash	Clear Puzzle 5	10 Medals
07	20,000 Ash	Clear Puzzle 6	16 Medals
08	25,000 Ash	Clear Puzzle 7	20 Medals
09	50,000 Ash	Clear Puzzle 8	30 Medals
10	100,000 Ash	Clear Puzzle 9	40 Medals
Credits	Clear Fun 1 to 10	Clear Puzzle 10	Clear Challenge 1 to 10
11	200,000 Ash	Clear Puzzle 10	44 Medals

14	2,000,000 Ash	Clear Puzzle 13	56 Medals
15	5,000,000 Ash	Clear Puzzle 14	58 Medals
16	10,000,000 Ash	Clear Puzzle 15	60 Medals

Note: not all the Paths will go up to Stage 16. The numbers should be considered theoretical in these cases.

4.8.6 Next Quest Condition Messages

In the Hub, the next Quest in each path is displayed as a Stone block.

The following condition messages are used to display the conditions:



Path	Message to Display	# value
Fun	# Ash	Ash for next quest
Puzzle	Clear Puzzle #	Current Puzzle Quest number
Challenge	# Medals	Target medal total

from display, and the message bar disappears as burning any, or none, of them.

After burning, the corresponding alternative object will have to appear (or fall down from the sky).

The following table describes the options:

Option	Effect	Enable Message	Disable Message
SelectReminder	Show Select Reminder in Message Bar	Burn me to Hide Select Reminder	Burn me to Show Select Reminder
FieldNames	Show Field Names	Burn me to Enable Field Names	Burn me to Disable Field Names
AlternateControls	Changes to the alternative control scheme	Burn me for Alternate Controls	Burn me for Normal Controls
Louder	Turns up volume	Burn me for Louder Volume	–
Softer	Turns down volume	Burn me for Quieter Volume	



Component	Example	Explanation
Type	Fun01 or Puzzle01 or Challenge01	Denotes which Path the field list appears in, and which step along that path it represents.
Title	"Avalon 6"	Optional If there is a title, it is shown on the title page when this block is in the select slot.
Random	Random 6	Optional If there is a Random field, it means the actual fields to be played should be randomly selected from the fields in the field list. The number given is the number of fields to play e.g. Random 1 would mean 'play a single field from the field list'.
BOF Marker		Beginning of Field List marker
Field 1 Field 2 Field 3 Field 4 Field 5 Field 6 etc.	tourmaline.fsf or tourmaline.fsf "Tourmaline"	Filename for the fields, in the sequence they are to be played. (Any number of fields can be listed, but the normal is 6) Optional Extra If there is a section in quotes, it means this is the name of the field, and this should be displayed at the start of the field.
EOF Marker		End of file marker



5. Audio

5.1. Table of Sound Effects



Sound Effect	Situation	Notes
Jump	Sound made while ascending	Sound made after pressing jump, while fireball ascends (duration: slightly less than 1 second)
Airborne	Sound made while in the air	A relatively quiet noise that can be heard while the fireball is in the air, and that ceases when it lands.
Landing from Jump	Fireball lands on block or ground	When the fireball touches down from a jump (irrespective of material landed upon)
Burning Block	Leaf Block	SFX when blocks of this type burn
Burning Block	Wood Block	SFX when blocks of this type burn

Burning Block	Metal Block	SFX when blocks of this type burn
Melting Block	Plastic Block	SFX when blocks of this type melt
Melting Block	Metal Block	SFX when blocks of this type melt
Melting Block	Stone Block	SFX when blocks of this type melt
Heat Level Changed	Yellow Hot	Probably unused, unless Water blocks added
Heat Level Changed	Orange Hot	Plays when the player 'levels up' to Orange
Heat Level Changed	Red Hot	Plays when the player 'levels up' to Red
Heat Level Changed	Blue Hot	Plays when the player 'levels up' to Blue
Heat Level Changed	White Hot	Plays when the player 'levels up' to White
Slam	Yellow Hot	Sound effect for when a Yellow Hot Fireball slams – a burst of flame
Slam	Orange Hot	Sound effect for when a Yellow Hot Fireball slams – a conflagration
Slam	Red Hot	Sound effect for when a Yellow Hot Fireball slams – a loud fireburst
Slam	Blue Hot	Sound effect for when a Yellow Hot Fireball slams – a huge explosion
Slam	White Hot	Sound effect for when a Yellow Hot Fireball slams – a “nuclear explosion”
Begin Fanfare	Start of Level	Short fanfare, rising scale or audio effect
Restart Noise	Restart level	A swishing noise; just to underline the level has been reset
Ash Clear	Ash Target Hit	Short rising scale or audio effect
End Fanfare	Reach Exit	Very short victory fanfare or rising scale
Complete Fanfare	End of Field List	Short reward fanfare (no medal)
Bronze Fanfare	Earn Bronze Medal	Short reward fanfare (variant of above)
Silver Fanfare	Earn Silver Medal	Short reward fanfare (variant of above)
Gold Fanfare	Earn Gold Medal	Short reward fanfare (variant of above)



This section provides advice to field designers for *Fireball*. Note that field designers are encouraged to go beyond the templates described here. The templates define how one should start thinking about the fields, but the actual fields can be built any way the field designer chooses.

6.2. Stages

In order to ensure the player has time to learn how the game works without recourse to an explicit tutorial, there is a strict order to the stages of the game. The following table provides a summary:

Code	Stage #	Blocks that will Burn	Blocks that will Melt	Blocks that are Inert
L	0	Leaf	-	Stone
W	1	Wood	-	Stone
LW	2	Leaf, Wood	-	Stone
C	3	Leaf, Wood, Coal	-	Stone
P	4	Leaf, Wood	Plastic	Stone
CP	5	Leaf, Wood, Coal	Plastic	Stone
F	6	Leaf, Wood, Coal	Plastic	Fire
M	7	Leaf, Wood	Metal	Fire
PM	8	Leaf, Wood, Coal	Plastic, Metal	Fire
S	9	Leaf, Wood, Coal	Plastic, Metal, Stone	Fire



6.3. Fuses

An important concept in *Fireball* field design is that of *fuses*. A fuse is a set of objects which are designed to be automatically ignited by the player, thus causing remote effects. Fuses can be used to:

- Create easy Chains
- Set fire to distant points in the landscape
- Give the player a time limit – i.e. a fuse may set fire to a bridge, which the player must then cross before it burns out.

A fuse which begins with Leaf blocks will always be lit by the player. (The field may begin with the player on such a fuse). Other types of fuses may require the player to become hotter before they can be lit.

6.4. Fields

6.4.1 Field Types

The following are the broad different types of field:

This style of field is likely to be Easy, with a few of Mid difficulty.

- **Sprint:** fields in which the goal is at the end of an “assault course”. The player naturally gets hotter as progressing, and so damages the terrain that will get them to the goal.

This style of field is likely to be Hard, except when there are no fuses, when it might become Easy.

- **Hunt:** fields in which the goal is hidden by some means, and the player must find it.

This style of field is likely to be Easy, as the player will eventually succeed.

- **Maze:** fields in which the goal can be seen, but a maze of some kind blocks progress.

This style of field is likely to be Easy, unless fuses are used to create time limitations in which case it could become Hard.

6.4.2 Field Filenames

The following filename format is used for fields:

XXXXXXXXXX.XX.XXXX.fsf

The first ten characters are an informal description of the level.

The next two characters represent the Stage (see below).

The next four characters represent the difficulty, either:

- Easy
- Mid
- Hard
- Hell

So an example field filename might be:

Smallhouse.LW.easy.fsf

This allows anyone building field lists to know that the field designer intends this particular field to be easy to complete.

6.5. Field List Template

Each field list can follow the following broad pattern *in terms of the difficulty of levels included*. As ever, however, field designers are encouraged to extemporise!

In the early game:

- Field 1 = Easy and very simplistic
- Field 2 = Easy
- Field 3 = Mid with a trick that makes it Easy
- Field 4 = Easy
- Field 5 = Mid
- Field 6 = Easy and rewarding

This pattern is designed to contain a range of different difficulties, arranged so that the feel of the field list is that of a ‘journey’.

In the later game:

- Field 1 = Mid
- Field 2 = Easy
- Field 3 = Hard with a trick that makes it Easy



Try not to make the last field of a field list be Hard, unless it has a trick that makes it Easy.

Fields in the Challenge path may be harder than this pattern suggests, while fields in the Fun path should all be Easy.

7. Appendix I: Target Audience

7.1. Evangelist Clusters

Evangelist clusters represent sections of the audience for the game with the potential and intent to spread knowledge of the game to a wider audience. The evangelist clusters for the game would be:

- **H3 Wanderers (looking for a unique experience):** a “Hardcore” gaming cluster (based on self-assessment and number of games purchased), this cluster is mixed gender and samples many different games over the course of each year. Their interest in novelty should aid in reaching them with this product. About 20% of Specialist Press reviewers fall into this cluster – these should be targeted by the game’s PR drive.
- **H2 Managers (looking for systems & puzzles):** Logical puzzles of all kinds appeal to this cluster. The original yet intuitive mechanics of *Fireball*, coupled with the puzzle-oriented gameplay, will appeal to players preferring this play style. Although there are many H2 type players in the specialist press they are unlikely evangelists, as they tend mostly to review strategy games.



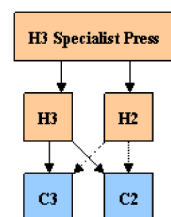
7.2. Target Clusters

Target clusters are the wider audience being targeted. The target clusters for the game would be:

- **C3 Wanderers:** related to the H3, these have similar game habits to the H3’s but play fewer games each year. They are attracted to novelty, and will buy into the game if it is recommended to them. If word-of-mouth for the game is significant, this cluster can drive game sales to levels above typical budget products.
- **C2 Manager:** a mass market cluster who can potentially be attracted by abstract games with a strong bias towards process orientation. They buy many games (but complete few), which makes them a likely target cluster for an unusual game at a budget price point.

7.3. Market Vector

The planned market vector for this game therefore consists of the following:



Orange squares represent sales primarily from hardcore gamers, whilst blue squares represent sales to a mass market audience.

Transition from H2 to C3 or C2 (dotted arrow) is most likely be a male H2 showing the game to friends, and therefore uptake from these vectors will be lower Conversely, transitions from H3 to C3 or C2 are more likely to be a friend making a recommendation for purchase, and therefore should help drive sales.

Core design incorporated from Concept version 1.0

8.2. Version 1.05

Changed references that read 'Combos' to saying 'Chains' (i.e. renamed 'Combos' to 'Chains').

Added a section about Chains.

Added a Template section giving advice to field designers.

8.3. Version 1.10

Added the concept of ignition time. A block has to have been on fire for a (short) amount of time (known as the ignition time) before it can set fire to other blocks.

Ignition times added to the block data table.

Added reference to the parameters for the fire effects. (Appendix I in this document).

Target Audience titled 'Appendix II'.



8.4. Version 1.15

Replaced the old conception of the menu system with the notion of a 'hyperobject' so that the main game engine can be used to power the menu system.

8.5. Version 1.20

Added the (already implied) functionality for Select – namely that pressing Select restarts the current level.

8.6. Version 1.25

Added 'Time Out' feature – if the player fails to complete a level in 20 minutes, the player automatically becomes White Hot (5) and can therefore automatically complete the level. However, none of the player's achievements will then be counted.

8.7. Version 1.5

Blocks are expressly sized at 4 m x 4 m x 4m, so talk of them being 1 m cube is incorrect. Updated in this version.

Jump heights are now 2 units, 4 units, 8 units, 12 units and 16 units.

8.8. Version 1.51

Specified: No block may burn at a temperature above its ignition temperature.

Removed some 'dead' material.

8.9. Version 1.52

Coal now ignites at Red Hot, not Orange Hot.

Revised burn times and ignition times:

Block Type	Block Colour	Melt	Burn	Burn Time	Ignition Time
1: Leaf	Green	No	1: Yellow Hot	10 seconds	0.1 seconds

4: Plastic	Orange	2: Orange Hot	3: Red Hot	15 seconds	1 second
5: Metal	Blue	4: Blue Hot	5: White Hot	90 seconds	1 second
6: Stone	Grey	5: White Hot	No	–	–
7: Fire	Red	No	No	–	–

8.10. Version 1.53

Minor edits.

Deleted the Time Out feature.



8.11. Version 1.54

Minor edits.

8.12. Version 1.6

Revised in line with the new structure.

Substantial revisions to Section 4, in line with the new structure.

Also, moved the M stages after F and before PM.

8.13. Version 1.75

Added Ribbons.

Added High Level state diagram and annotations.

Added Overlays

Added Options

Added sound effects table (from Pending Features).

Noted the new name 'Quest' as a synonym for field list.

8.14. Version 1.76

Added 'Hell' difficulty.

8.15. Version 1.8

Revised flow diagrams to include:

- Language select screen
- Pause state
- Added Continue/Quit overlay to Pause screen.

8.16. Version 1.9

Incorporated changes made from Update.doc, and removed sections no longer relevant.

This included changes to Hub display and howQuests were unlocked.

Get daily news, dev blogs, and stories from Game Developer straight to your inbox

Subscribe

LATEST JOBS

Treyarch

PLAYA VISTA, LOS ANGELES, CA, USA

2.03.23

High Moon Studios

CARLSBAD, CA, USA

2.03.23

SENIOR PHYSICS GAMEPLAY ENGINEER – TREYARCH

VFX ARTIST – HIGH MOON STUDIOS

Anne Arundel Community College

ARNOLD, MD, USA

1.30.23

Treyarch Vancouver

VANCOUVER, BC, CANADA

2.02.23

INSTRUCTOR/ASSISTANT PROFESSOR, GAME ART

PRODUCER – TREYARCH VANCOUVER

MORE JOBS 

CONNECT WITH US

Explore the

Subscribe to

Follow us

Browse open positions across the game industry or recruit new talent for your studio

[BROWSE](#)

Get daily Game Developer top stories every morning straight into your inbox

[SUBSCRIBE](#)

Follow us @gameadevaotcom to stay up-to-date with the latest news & insider information about events & more



Discover More From Informa Tech

[Game Developers Conference](#)

[Game Developer Jobs](#)

[GDC Vault](#)

[Game Career Guide](#)

[Independent Games Festival](#)

[Game Developers Choice Awards](#)

[Omdia](#)

Working With Us

[Contact Us](#)

[About Us](#)

[Advertise](#)

Follow Game Developer On Social



[Home](#) [Cookies](#) [Privacy](#) [Terms](#)